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Code JupyterLab Python 3 (ipykernel)

[1]:
```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

PART A

[2]:
```python
df = pd.read_csv("DataSets/employee_performance_dataset.csv")
df.head()
```

[2]:

	Employee_ID	Name	Department	Age	Gender	Years_at_Company	Goals_Achieved	Projects_Completed	Trainings_Attended	Attendance_%	Customer_Feedback
0	E1000	Seth	Sales	50	Male	11	12	5	1	74.679836	8
1	E1001	Maria	IT	45	Male	3	10	5	1	91.659963	6
2	E1002	Mary	HR	42	Male	1	16	10	5	70.211989	9
3	E1003	Kyle	Sales	49	Male	12	20	3	3	85.427033	9
4	E1004	Sandra	Marketing	28	Female	5	13	7	1	98.466566	2

[5]:
```python
df.isna().sum()
```

[5]:
```
Employee_ID          0
Name                 0
Department           0
Age                  0
Gender               0
Years_at_Company     0
Goals_Achieved       0
Projects_Completed   0
Trainings_Attended   0
Attendance_%         0
Customer_Feedback    0
Salary(LPA)          0
```

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Code JupyterLab Python 3 (ipykernel)

PART B

[16]:
```python
df.groupby("Department")["Performance_Score"].mean()
df
```

[16]:

	Employee_ID	Name	Department	Age	Gender	Years_at_Company	Goals_Achieved	Projects_Completed	Trainings_Attended	Attendance_%	Customer_Feedbac
0	E1000	Seth	Sales	50	Male	11	12	5	1	74.679836	
1	E1001	Maria	IT	45	Male	3	10	5	1	91.659963	
2	E1002	Mary	HR	42	Male	1	16	10	5	70.211989	
3	E1003	Kyle	Sales	49	Male	12	20	3	3	85.427033	
4	E1004	Sandra	Marketing	28	Female	5	13	7	1	98.466566	
...	...	...	...	...	...	...	...	...	...	...	...
495	E1495	James	IT	57	Female	3	5	10	3	91.771777	1
496	E1496	Jesse	IT	27	Male	6	5	10	1	92.284749	
497	E1497	Sean	Finance	42	Female	9	16	4	4	85.381792	1
498	E1498	Nicholas	Marketing	49	Male	1	15	9	0	77.260662	
499	E1499	Darren	Sales	44	Female	4	19	1	0	70.474396	

500 rows × 13 columns

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Code JupyterLab Python 3 (ipykernel)

[21]:
```python
df.sort_values(by = "Performance_Score",ascending = False).head(10)
```

[21]:

	Employee_ID	Name	Department	Age	Gender	Years_at_Company	Goals_Achieved	Projects_Completed	Trainings_Attended	Attendance_%	Customer_Feedback
302	E1302	Carol	Sales	23	Male	9	20	9	3	85.724775	10
166	E1166	John	HR	39	Female	1	20	10	3	76.988784	7
245	E1245	Brittany	Marketing	32	Female	7	19	10	4	75.537554	8
252	E1252	Lisa	HR	49	Female	7	20	10	0	81.727180	9
467	E1467	Patrick	Sales	57	Female	7	20	8	5	81.795856	5
154	E1154	Kelly	Marketing	55	Female	7	19	8	5	74.050038	8
342	E1342	Scott	Finance	32	Female	2	19	9	4	77.629659	6
81	E1081	Martin	IT	22	Female	5	20	7	5	90.751432	5
202	E1202	Sarah	IT	52	Female	6	20	9	0	98.579459	9
28	E1028	Luke	Sales	37	Male	10	20	9	3	85.889517	2

[22]:
```python
df["Salary(LPA)"].mean()
```

[22]:
```
np.float64(11.563600000000001)
```

[24]:
```python
df.groupby("Department")["Salary(LPA)"].mean()
```

[24]:
```
Department
Finance      11.619423
HR           10.774149
IT           12.030306
Marketing    11.666875
Sales        11.684239
Name: Salary(LPA), dtype: float64
```

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Code JupyterLab Python 3 (ipykernel)

[25]:
```
df.groupby("Gender")["Performance_Score"].mean()
```

[25]:
```
Gender
Female    7.528629
Male      7.542460
Name: Performance_Score, dtype: float64
```

[49]:
```
cor1 = df[["Attendance_%","Customer_Feedback","Trainings_Attended","Salary(LPA)","Performance_Score"]].corr().round(2)
cor1
```

[49]:

	Attendance_%	Customer_Feedback	Trainings_Attended	Salary(LPA)	Performance_Score
Attendance_%	1.00	0.03	0.08	-0.03	-0.02
Customer_Feedback	0.03	1.00	-0.05	0.03	0.13
Trainings_Attended	0.08	-0.05	1.00	0.00	0.17
Salary(LPA)	-0.03	0.03	0.00	1.00	-0.00
Performance_Score	-0.02	0.13	0.17	-0.00	1.00

PART C

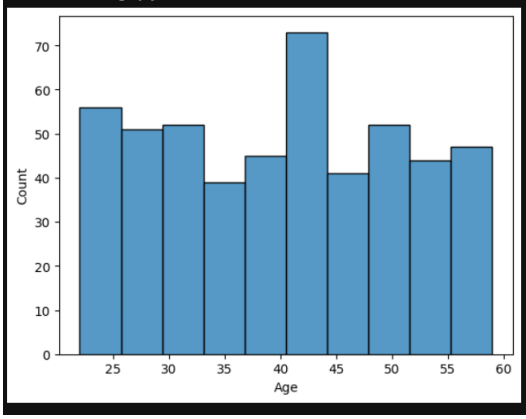
[30]:
```
avg = df.groupby("Department")["Performance_Score"].mean()
avg
```

[30]:
```
Department
Finance      7.658654
HR           7.073404
IT           7.702041
Marketing    7.535714
Sales        7.691304
Name: Performance_Score, dtype: float64
```

[36]:
```
sns.barplot(x = "Department",y = "Performance_Score"   ,data = df)
```

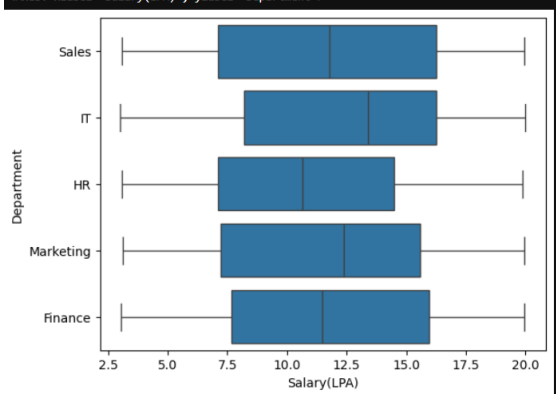
```
sns.histplot(x = "Age",data = df)
```

```
<Axes: xlabel='Age', ylabel='Count'>
```



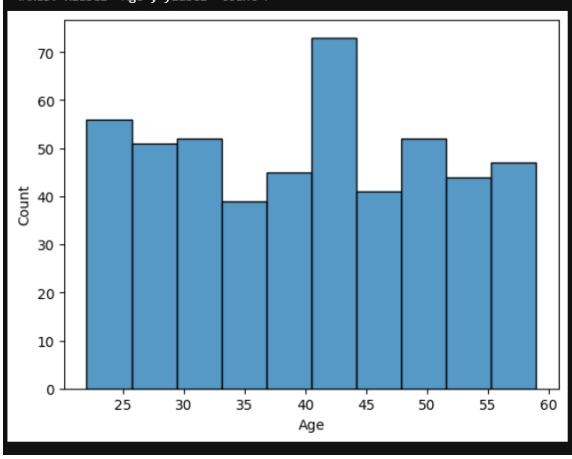

```
sns.boxplot(x = "Salary(LPA)",y = "Department"   ,data = df)
```

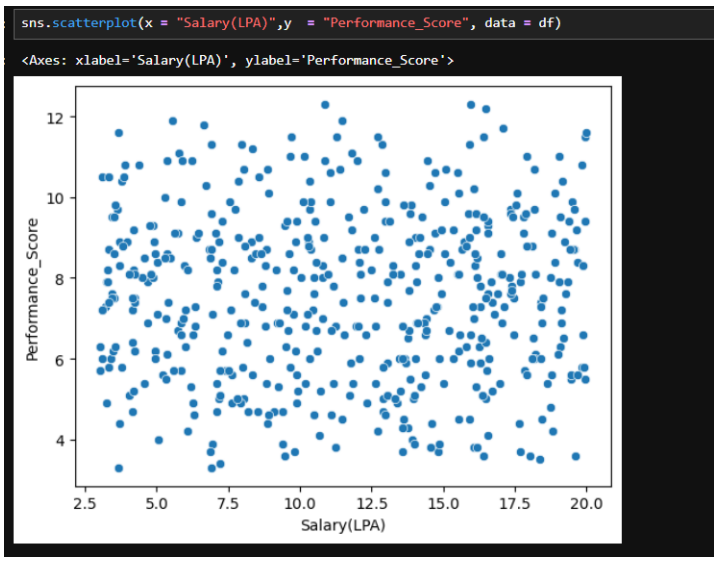
```
<Axes: xlabel='Salary(LPA)', ylabel='Department'>
```

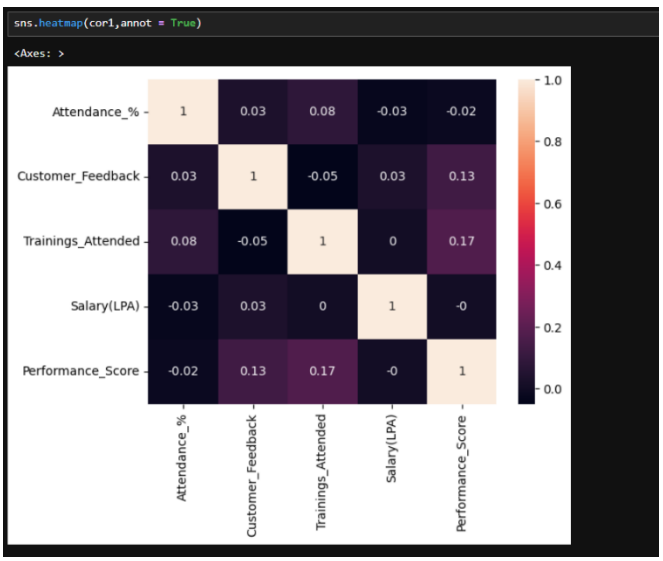



```
sns.histplot(x = "Age",data = df)
```

```
<Axes: xlabel='Age', ylabel='Count'>
```







Employee Performance & HR Analytics – Data Analysis Case Study

Part A: Data Preparation

Part B: Exploratory Data Analysis (EDA)

Part C: Visualization (Matplotlib + Seaborn)

Expected Outcomes

- **Which department performs best on average?**
- **Who are the top performers in the company?**
- **Do men and women differ in performance?**
- **Does higher salary mean higher performance?**
- **Which factors correlate strongly with performance?**